

SATYENDRA

ROBOTICS ENGINEER | MOTION PLANNING | CONTROLS | SYSTEM INTEGRATION

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Portfolio: sahnisatyendra022-cloud.github.io/portfolio-page/#about

PROFESSIONAL SUMMARY

Robotics Engineer with 3+ years of hands-on robotics development experience across robot motion control, ROS 2, embedded systems, simulation, perception, and system integration. Experience developing ROS 2-based robotic applications, robot kinematics and trajectory planning, vision-guided automation, and autonomous navigation workflows. Worked across software, electronics, firmware, vision, and mechanical integration, including end-to-end technical ownership of a vision-guided Delta robot system. Strong experience with C++, Python, Linux, Isaac Sim, embedded controllers, sensors, and motor-control systems.

CORE TECHNICAL SKILLS

Robotics & ROS: ROS 2 (Humble/Jazzy), Nav2, RTAB-Map, MoveIt 2, TF/URDF, SLAM, localization, motion planning, kinematics, trajectory planning

Programming: C++, Python, Embedded C, Linux, Git

Simulation: NVIDIA Isaac Sim, Isaac Lab, PyBullet, Gazebo, PPO / reinforcement learning

Controls & Embedded: Teensy 4.1, STM32, ESP32/ESP8266, Raspberry Pi, PWM, PID, motor/servo control, FOC concepts

Perception: LiDAR, RGB-D cameras, IMU, encoders, OpenCV, YOLOv8, MediaPipe, camera-to-robot calibration, sensor fusion

Hardware & Communication: Motor/servo drivers, PCB/control-panel integration, UART, I2C, SPI, CAN, RS-485/Modbus, BLE, MQTT/HTTP

Tools: SolidWorks, STM32CubeIDE/CubeMX, Keil, Proteus, easyEDA, ST-Link

PROFESSIONAL EXPERIENCE

Robotics Engineer — WeVois Labs Pvt. Ltd. *Feb 2026 – Present*

- Technical owner of a high-speed vision-guided Delta robot pick-and-place system, covering architecture, firmware, electronics, vision integration, and deployment.
- Developed Teensy 4.1 motion-control firmware with custom Delta inverse kinematics, synchronized multi-axis control, fault recovery, and safety sequencing.
- Designed the controller circuit and PCB, handled assembly/control-panel wiring, and developed a PC-side SDK for motion commands, calibration, monitoring, and control.
- Integrated object detection, camera-to-robot calibration, and conveyor tracking; worked with mechanical and vision teams during system integration and migrated the prototype from stepper motors to AC servos.

Robotics Engineer — HighFlow Industries Pvt. Ltd. (NextStep Robotics) *Dec 2024 – Feb 2026*

- Developed ROS 2 workflows for autonomous mobile robotics, including mapping, localization, navigation, TF/URDF, and sensor-data integration.
- Built the AMR URDF and simulation environment and integrated RTAB-Map, Nav2, and simulated 3D LiDAR data for mapping and navigation.
- Validated the AMR mapping and navigation stack in NVIDIA Isaac Sim; real-robot hardware validation with 3D LiDAR/cameras was outside the project scope.

Embedded & Robotics R&D Consultant / Contract — XPRT Enforcement Pvt. Ltd. *Aug 2022 – Feb 2023*

- Integrated GPS/radar telematics for real-time vehicle tracking.
- Authored technical architecture documentation for municipal IoT deployment and supported hardware/software integration.

Embedded Engineer — Experio Traffiko System Pvt. Ltd. *Nov 2020 – Oct 2021*

- Developed embedded hardware for Speedo Soft, an AI-driven traffic-enforcement system using ANPR-based violation detection, including circuit design, PCB layout, and SMD assembly.
- Developed firmware for radar-based speed detection, dual-battery intelligent power management, and breath-analyzer modules; handled hardware development from schematic through PCB validation.

SELECTED ROBOTICS PROJECTS

- **Vision-Guided Delta Robot** — Custom Delta IK, synchronized multi-axis control, vision calibration, conveyor tracking, servo migration, and end-to-end system integration.
- **Autonomous AMR** — ROS 2, RTAB-Map, Nav2, URDF, simulated 3D LiDAR, and Isaac Sim-based mapping/navigation validation.
- **RL-Based Robotic Arm** — PPO training in Isaac Sim/Isaac Lab with custom reward/observation logic and ROS 2 validation.
- **Vision-Based Pick & Place** — YOLO/OpenCV detection, camera-to-robot calibration, inverse kinematics, and trajectory planning.
- **Humanoid Robotic Hand** — Multi-DOF finger/arm control using IMU/vision input with a synchronized simulation digital twin.
- **QDD Actuator & Biped Simulation** — BLDC/planetary QDD actuator evaluation and gait/trajectory control in PyBullet using IK and PID/MPC concepts.

EDUCATION

- B.Tech, Electronics & Communication Engineering — Dr. A.P.J. Abdul Kalam Technical University | 2021 – 2024
- Diploma, Electronics Engineering — Board of Technical Education, Uttar Pradesh | 2018 – 2020

TRAINING

IoT Training, Softpro India (2024) | Python with IoT & Raspberry Pi, Softpro India (2020) | PCB Design, Softpro India (2020) | Embedded Systems & OS, ITI Limited Mankapur (2019)